

AquaScope — Project Proposal

1. Summary

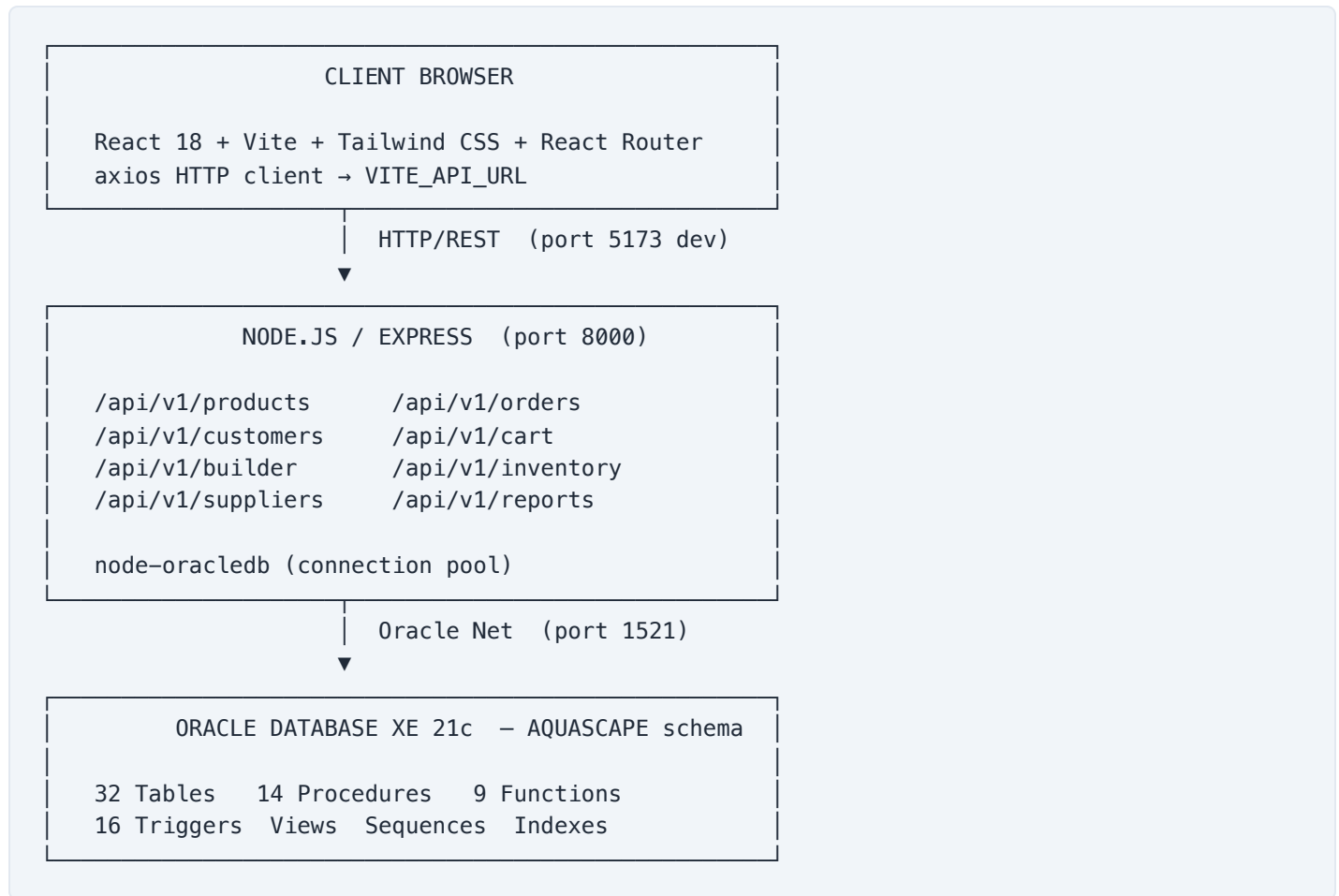
AquaScope is a full-stack e-commerce and management platform for an aquarium retail business. It supports two user groups: **staff/admin** (who manage products, inventory, orders, and suppliers) and **customers** (who browse the shop, build custom aquarium setups, and place orders).

The system is built on an Oracle Database backend with a Node.js/Express API layer and a React frontend. All critical business logic — order placement, inventory updates, aquarium validation, and supplier workflows — is implemented as Oracle PL/SQL stored procedures and functions, ensuring data integrity at the database level.

2. Technology Stack

Layer	Technology	Version
Database	Oracle Database Express Edition	21.3.0-XE
ORM / Driver	node-oracledb	6.6.0
Backend Framework	Express.js	4.21.0
Backend Runtime	Node.js	20 (LTS)
Frontend Framework	React	18.3.1
Frontend Router	React Router DOM	6.26.0
Build Tool	Vite	5.4.8
CSS Framework	Tailwind CSS	3.4.13
HTTP Client	Axios	1.7.7
Charts	Recharts	2.12.7
Icons	Lucide React	0.447.0
Containerisation	Docker / Docker Compose	—
Environment Config	dotenv	16.4.5

3. System Architecture



All three tiers run as Docker containers orchestrated by `docker-compose.yml`.

4. Features

4.1 Product Catalogue

- Five product types: **Fish, Plant, Tank, Equipment, Decoration**
- Per-product fields: SKU, unit price, cost price, image URL, category, active flag
- Type-specific attributes: species data for fish, light requirements for plants, volume and dimensions for tanks, power and suitability for equipment

4.2 Inventory Management

- Real-time stock tracking with `qty_on_hand` and `qty_reserved` separation
- Configurable reorder level and reorder quantity per product
- Automatic **low-stock alerts** generated by trigger when stock falls at or below reorder level
- Alert acknowledgment workflow (resolve with user/timestamp)

- Immutable **inventory movement ledger** — every stock change (SALE, PURCHASE, RETURN, ADJUSTMENT, DAMAGE) is recorded with `quantity_delta` and `qty_after` snapshot

4.3 Aquarium Builder

- Customers design custom aquarium setups by selecting a tank and adding fish, plants, equipment, and decorations
- Three server-side **validation functions** run on save:
 - **Capacity validation** — fish stocking density vs tank volume
 - **Water type validation** — all items must match setup water type (FRESHWATER / SALTWATER / BRACKISH)
 - **Temperature validation** — fish temperature ranges must overlap
- **Compatibility rules** system — admin-defined INCOMPATIBLE / REQUIRES / NEUTRAL rules between products
- Setups can be saved as public share links or converted directly to orders

4.4 Shopping Cart & Orders

- One persistent cart per customer
- Order placement via `place_order` procedure — full ACID transaction covering:
 - Stock availability check
 - Discount code validation (date range, usage limit, minimum order amount)
 - Tax calculation (6%)
 - Cart-to-order conversion
 - Inventory deduction
 - Cart clearance
- Order status state machine: `PENDING → CONFIRMED → PROCESSING → SHIPPED → DELIVERED` (or `CANCELLED` / `REFUNDED`)
- Payment tracking per order (CASH, CARD, BANK_TRANSFER, E_WALLET)
- Return request and processing workflow with selective restocking of GOOD-condition items

4.5 Supplier & Purchase Orders

- Supplier directory with contact details and payment terms
- Purchase order workflow: `DRAFT → SUBMITTED → APPROVED → SHIPPED → RECEIVED`
- PO receiving records actual quantities delivered vs ordered
- Batch traceability on received stock (expiry date, cost, quantity)
- Receiving automatically calls `update_inventory` (move_type = PURCHASE)

4.6 Customer Management

- Customer profiles with contact details
- Lifetime value calculated via aggregation over order history
- Per-customer order history and return history

4.7 Reporting & Analytics

- **Dashboard KPIs:** total revenue, orders today, low-stock count, active customers
- **Monthly sales** chart (revenue, units, average order value)
- **Best sellers** by revenue and by units sold
- **Profit analysis** by product (revenue, COGS, gross margin %)
- **Fast movers:** 30-day sales velocity for demand planning
- **Audit log** viewer — every INSERT, UPDATE, DELETE on core tables
- **Stock movement log** — detailed ledger with reference tracking
- **Oracle objects browser** — live inventory of all DB objects (tables, views, procedures, triggers, functions)

4.8 Admin Tooling

- Compatibility rule management
- Inventory adjustment (manual stock correction with reason)
- Role-based access structure (ADMIN, STAFF, CUSTOMER)

5. Database Schema

The schema is divided across four SQL files representing four functional domains.

5.1 Core Domain (`01_core_schema.sql`)

ROLES

Column	Type	Constraints
role_id	NUMBER	PK, seq_role
role_name	VARCHAR2(50)	UNIQUE NOT NULL
description	VARCHAR2(200)	

USERS

Column	Type	Constraints
user_id	NUMBER	PK, seq_user
username	VARCHAR2(100)	UNIQUE NOT NULL
password_hash	VARCHAR2(255)	NOT NULL
email	VARCHAR2(200)	UNIQUE NOT NULL
full_name	VARCHAR2(200)	
role_id	NUMBER	FK → ROLES
is_active	NUMBER(1)	DEFAULT 1
last_login	TIMESTAMP	
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

CUSTOMERS

Column	Type	Constraints
customer_id	NUMBER	PK, seq_customer
first_name	VARCHAR2(100)	NOT NULL
last_name	VARCHAR2(100)	NOT NULL
email	VARCHAR2(200)	UNIQUE NOT NULL
phone	VARCHAR2(20)	
address	VARCHAR2(500)	
city	VARCHAR2(100)	
country	VARCHAR2(100)	DEFAULT 'Malaysia'
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

SUPPLIERS

Column	Type	Constraints
supplier_id	NUMBER	PK, seq_supplier
supplier_name	VARCHAR2(200)	NOT NULL
contact_person	VARCHAR2(200)	
email	VARCHAR2(200)	
phone	VARCHAR2(20)	
address	VARCHAR2(500)	
payment_terms	VARCHAR2(50)	CHECK IN ('NET30','NET60','COD','PREPAID')
is_active	NUMBER(1)	DEFAULT 1
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

CATEGORIES

Column	Type	Constraints
category_id	NUMBER	PK, seq_category
category_name	VARCHAR2(100)	NOT NULL
parent_id	NUMBER	FK → CATEGORIES (self-ref)
description	VARCHAR2(500)	

PRODUCTS

Column	Type	Constraints
product_id	NUMBER	PK, seq_product
product_name	VARCHAR2(200)	NOT NULL
product_type	VARCHAR2(20)	CHECK IN ('FISH','PLANT','TANK','EQUIPMENT','DECORATION')
category_id	NUMBER	FK → CATEGORIES
sku	VARCHAR2(50)	UNIQUE
unit_price	NUMBER(10,2)	NOT NULL
cost_price	NUMBER(10,2)	
description	VARCHAR2(1000)	
image_url	VARCHAR2(500)	
is_active	NUMBER(1)	DEFAULT 1
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

INVENTORY

Column	Type	Constraints
inventory_id	NUMBER	PK, seq_inventory
product_id	NUMBER	FK → PRODUCTS UNIQUE
qty_on_hand	NUMBER	DEFAULT 0 NOT NULL
qty_reserved	NUMBER	DEFAULT 0
reorder_level	NUMBER	DEFAULT 10
reorder_qty	NUMBER	DEFAULT 50
last_updated	TIMESTAMP	DEFAULT SYSTIMESTAMP

TANKS

Column	Type	Constraints
tank_id	NUMBER	PK, seq_tank
product_id	NUMBER	FK → PRODUCTS
volume_ liters	NUMBER(8,2)	
length_cm	NUMBER(6,2)	
width_cm	NUMBER(6,2)	
height_cm	NUMBER(6,2)	
material	VARCHAR2(20)	CHECK IN ('GLASS','ACRYLIC')
shape	VARCHAR2(30)	CHECK IN ('RECTANGULAR','BOWFRONT','HEX','CUBE','CYLINDRICAL','CORNER','L-SHAPE')
has_hood	NUMBER(1)	DEFAULT 0

FISH

Column	Type	Constraints
fish_id	NUMBER	PK, seq_fish
product_id	NUMBER	FK → PRODUCTS
species_name	VARCHAR2(200)	
water_type	VARCHAR2(20)	CHECK IN ('FRESHWATER', 'SALTWATER', 'BRACKISH')
min_temp_c	NUMBER(4,1)	
max_temp_c	NUMBER(4,1)	
min_ph	NUMBER(4,2)	
max_ph	NUMBER(4,2)	
max_size_cm	NUMBER(5,1)	
stocking_density	NUMBER(5,2)	litres per fish
care_level	VARCHAR2(10)	CHECK IN ('EASY', 'MEDIUM', 'HARD')
is_aggressive	NUMBER(1)	DEFAULT 0
diet	VARCHAR2(100)	

PLANTS

Column	Type	Constraints
plant_id	NUMBER	PK, seq_plant
product_id	NUMBER	FK → PRODUCTS
species_name	VARCHAR2(200)	
water_type	VARCHAR2(20)	CHECK IN ('FRESHWATER', 'SALTWATER', 'BRACKISH')
light_requirement	VARCHAR2(10)	CHECK IN ('LOW', 'MEDIUM', 'HIGH')
co2_required	NUMBER(1)	DEFAULT 0
growth_rate	VARCHAR2(10)	CHECK IN ('SLOW', 'MEDIUM', 'FAST')
placement	VARCHAR2(20)	CHECK IN ('FOREGROUND', 'MIDGROUND', 'BACKGROUND', 'FLOATING')

EQUIPMENT

Column	Type	Constraints
equipment_id	NUMBER	PK, seq_equipment
product_id	NUMBER	FK → PRODUCTS
equipment_type	VARCHAR2(30)	CHECK IN ('FILTER','HEATER','LIGHTING','AIR_PUMP','CO2_SYSTEM','UV_STERILIZER','PROTEIN_SKIMMER','WAVE_MAKER')
power_watts	NUMBER(8,2)	
suitable_liters_min	NUMBER(8,2)	
suitable_liters_max	NUMBER(8,2)	

DECORATIONS

Column	Type	Constraints
deco_id	NUMBER	PK, seq_deco
product_id	NUMBER	FK → PRODUCTS
deco_type	VARCHAR2(30)	CHECK IN ('ROCK','DRIFTWOOD','CASTLE','CAVE','PLANT_DECORATION','SUBSTRATE','CORAL','STATUE')
material	VARCHAR2(100)	
is_natural	NUMBER(1)	DEFAULT 0
safe_water_type	VARCHAR2(20)	CHECK IN ('FRESHWATER','SALTWATER','BOTH')

5.2 Aquarium Builder Domain (02_aquarium_schema.sql)

AQUARIUM_SETUPS

Column	Type	Constraints
setup_id	NUMBER	PK, seq_setup
customer_id	NUMBER	FK → CUSTOMERS
tank_id	NUMBER	FK → TANKS
setup_name	VARCHAR2(200)	NOT NULL
water_type	VARCHAR2(20)	CHECK IN ('FRESHWATER','SALTWATER','BRACKISH')
status	VARCHAR2(20)	DEFAULT 'DRAFT', CHECK IN ('DRAFT','SAVED','ORDERED','ARCHIVED')
target_temp_c	NUMBER(4,1)	
target_ph	NUMBER(4,2)	
description	VARCHAR2(1000)	
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP
updated_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

SETUP_ITEMS

Column	Type	Constraints
setup_item_id	NUMBER	PK, seq_setup_item
setup_id	NUMBER	FK → AQUARIUM_SETUPS ON DELETE CASCADE
product_id	NUMBER	FK → PRODUCTS
item_type	VARCHAR2(20)	NOT NULL
quantity	NUMBER	DEFAULT 1
notes	VARCHAR2(500)	

COMPATIBILITY_RULES

Column	Type	Constraints
rule_id	NUMBER	PK, seq_compat_rule
product_id_a	NUMBER	FK → PRODUCTS
product_id_b	NUMBER	FK → PRODUCTS
rule_type	VARCHAR2(20)	CHECK IN ('INCOMPATIBLE','REQUIRES','NEUTRAL')
severity	VARCHAR2(10)	CHECK IN ('WARNING','ERROR')
reason	VARCHAR2(500)	
created_by	NUMBER	FK → USERS
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

SAVED_SETUPS

Column	Type	Constraints
saved_id	NUMBER	PK, seq_saved_setup
setup_id	NUMBER	FK → AQUARIUM_SETUPS
customer_id	NUMBER	FK → CUSTOMERS
share_code	VARCHAR2(20)	UNIQUE
is_public	NUMBER(1)	DEFAULT 0
saved_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

WISHLIST

Column	Type	Constraints
wishlist_id	NUMBER	PK
customer_id	NUMBER	FK → CUSTOMERS
product_id	NUMBER	FK → PRODUCTS
added_at	TIMESTAMP	DEFAULT SYSTIMESTAMP
		UNIQUE (customer_id, product_id)

CART

Column	Type	Constraints
cart_id	NUMBER	PK, seq_cart
customer_id	NUMBER	FK → CUSTOMERS UNIQUE
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP
updated_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

CART_ITEMS

Column	Type	Constraints
cart_item_id	NUMBER	PK, seq_cart_item
cart_id	NUMBER	FK → CART ON DELETE CASCADE
product_id	NUMBER	FK → PRODUCTS
quantity	NUMBER	DEFAULT 1 NOT NULL
added_at	TIMESTAMP	DEFAULT SYSTIMESTAMP
		UNIQUE (cart_id, product_id)

5.3 Order Domain (03_order_schema.sql)

DISCOUNTS

Column	Type	Constraints
discount_id	NUMBER	PK, seq_discount
code	VARCHAR2(50)	UNIQUE NOT NULL
discount_type	VARCHAR2(10)	CHECK IN ('PERCENT','FIXED')
value	NUMBER(10,2)	NOT NULL
min_order_amt	NUMBER(10,2)	DEFAULT 0
max_uses	NUMBER	NULL = unlimited
used_count	NUMBER	DEFAULT 0
is_active	NUMBER(1)	DEFAULT 1
valid_from	DATE	
valid_until	DATE	
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

ORDERS

Column	Type	Constraints
order_id	NUMBER	PK, seq_order (starts at 1000)
customer_id	NUMBER	FK → CUSTOMERS
setup_id	NUMBER	FK → AQUARIUM_SETUPS (nullable)
status	VARCHAR2(20)	DEFAULT 'PENDING', CHECK IN state machine
subtotal	NUMBER(12,2)	
discount_total	NUMBER(12,2)	DEFAULT 0
tax_amount	NUMBER(12,2)	DEFAULT 0
total_amount	NUMBER(12,2)	
shipping_address	VARCHAR2(500)	
notes	VARCHAR2(1000)	
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP
updated_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

ORDER_ITEMS

Column	Type	Constraints
order_item_id	NUMBER	PK, seq_order_item
order_id	NUMBER	FK → ORDERS
product_id	NUMBER	FK → PRODUCTS
quantity	NUMBER	NOT NULL
unit_price	NUMBER(10,2)	price snapshot at order time
line_total	NUMBER(12,2)	

ORDER_DISCOUNTS

Column	Type	Constraints
order_id	NUMBER	FK → ORDERS
discount_id	NUMBER	FK → DISCOUNTS
amount_saved	NUMBER(10,2)	
		PK (order_id, discount_id)

PAYMENTS

Column	Type	Constraints
payment_id	NUMBER	PK
order_id	NUMBER	FK → ORDERS
amount	NUMBER(12,2)	NOT NULL
payment_method	VARCHAR2(20)	CHECK IN ('CASH','CARD','BANK_TRANSFER','E_WALLET')
payment_status	VARCHAR2(20)	CHECK IN ('PENDING','COMPLETED','FAILED','REFUNDED')
transaction_ref	VARCHAR2(100)	
paid_at	TIMESTAMP	
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

RETURNS

Column	Type	Constraints
return_id	NUMBER	PK, seq_return
order_id	NUMBER	FK → ORDERS
customer_id	NUMBER	FK → CUSTOMERS
return_status	VARCHAR2(20)	CHECK IN ('REQUESTED','APPROVED','REJECTED','REFUNDED')
reason	VARCHAR2(500)	
refund_amount	NUMBER(12,2)	
requested_at	TIMESTAMP	DEFAULT SYSTIMESTAMP
processed_at	TIMESTAMP	
processed_by	NUMBER	FK → USERS

RETURN_ITEMS

Column	Type	Constraints
return_item_id	NUMBER	PK, seq_return_item
return_id	NUMBER	FK → RETURNS
order_item_id	NUMBER	FK → ORDER_ITEMS
product_id	NUMBER	FK → PRODUCTS
quantity	NUMBER	NOT NULL
condition_code	VARCHAR2(20)	CHECK IN ('GOOD','DAMAGED','DEAD','OTHER')
restock	NUMBER(1)	DEFAULT 0

5.4 Supplier & Inventory Domain (04_supplier_schema.sql)

SUPPLIER_PO

Column	Type	Constraints
po_id	NUMBER	PK, seq_spo (starts at 5000)
supplier_id	NUMBER	FK → SUPPLIERS
created_by	NUMBER	FK → USERS
po_status	VARCHAR2(20)	DEFAULT 'DRAFT', state machine
total_amount	NUMBER(12,2)	DEFAULT 0
expected_date	DATE	
received_date	DATE	
notes	VARCHAR2(1000)	
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

SUPPLIER_PO_ITEMS

Column	Type	Constraints
po_item_id	NUMBER	PK, seq_spo_item
po_id	NUMBER	FK → SUPPLIER_PO
product_id	NUMBER	FK → PRODUCTS
quantity_ord	NUMBER	NOT NULL
quantity_recv	NUMBER	DEFAULT 0
unit_cost	NUMBER(10,2)	NOT NULL
line_total	NUMBER(12,2)	

STOCK_BATCHES

Column	Type	Constraints
batch_id	NUMBER	PK, seq_batch
product_id	NUMBER	FK → PRODUCTS
po_id	NUMBER	FK → SUPPLIER_PO
quantity	NUMBER	NOT NULL
unit_cost	NUMBER(10,2)	
received_date	DATE	
expiry_date	DATE	
batch_notes	VARCHAR2(500)	

LOW_STOCK_ALERTS

Column	Type	Constraints
alert_id	NUMBER	PK, seq_alert
product_id	NUMBER	FK → PRODUCTS
current_qty	NUMBER	
reorder_level	NUMBER	
is_resolved	NUMBER(1)	DEFAULT 0
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP
resolved_at	TIMESTAMP	
resolved_by	NUMBER	FK → USERS

INVENTORY_MOVEMENTS

Column	Type	Constraints
movement_id	NUMBER	PK, seq_inv_move
product_id	NUMBER	FK → PRODUCTS
movement_type	VARCHAR2(20)	CHECK IN ('SALE','PURCHASE','RETURN','ADJUSTMENT','DAMAGE','RESERVED','UNRES
quantity_delta	NUMBER	positive = in, negative = out
qty_after	NUMBER	snapshot of qty_on_hand after move
reference_id	NUMBER	linked order_id / po_id / return_id
reference_type	VARCHAR2(20)	CHECK IN ('ORDER','PO','RETURN','ADJUSTMENT','SETUP')
notes	VARCHAR2(500)	
created_by	NUMBER	FK → USERS
created_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

AUDIT_LOG

Column	Type	Constraints
audit_id	NUMBER	PK, IDENTITY
table_name	VARCHAR2(100)	
record_id	NUMBER	
operation	VARCHAR2(10)	CHECK IN ('INSERT','UPDATE','DELETE')
old_values	CLOB	JSON snapshot
new_values	CLOB	JSON snapshot
changed_by	NUMBER	FK → USERS
changed_at	TIMESTAMP	DEFAULT SYSTIMESTAMP

6. ER Diagram

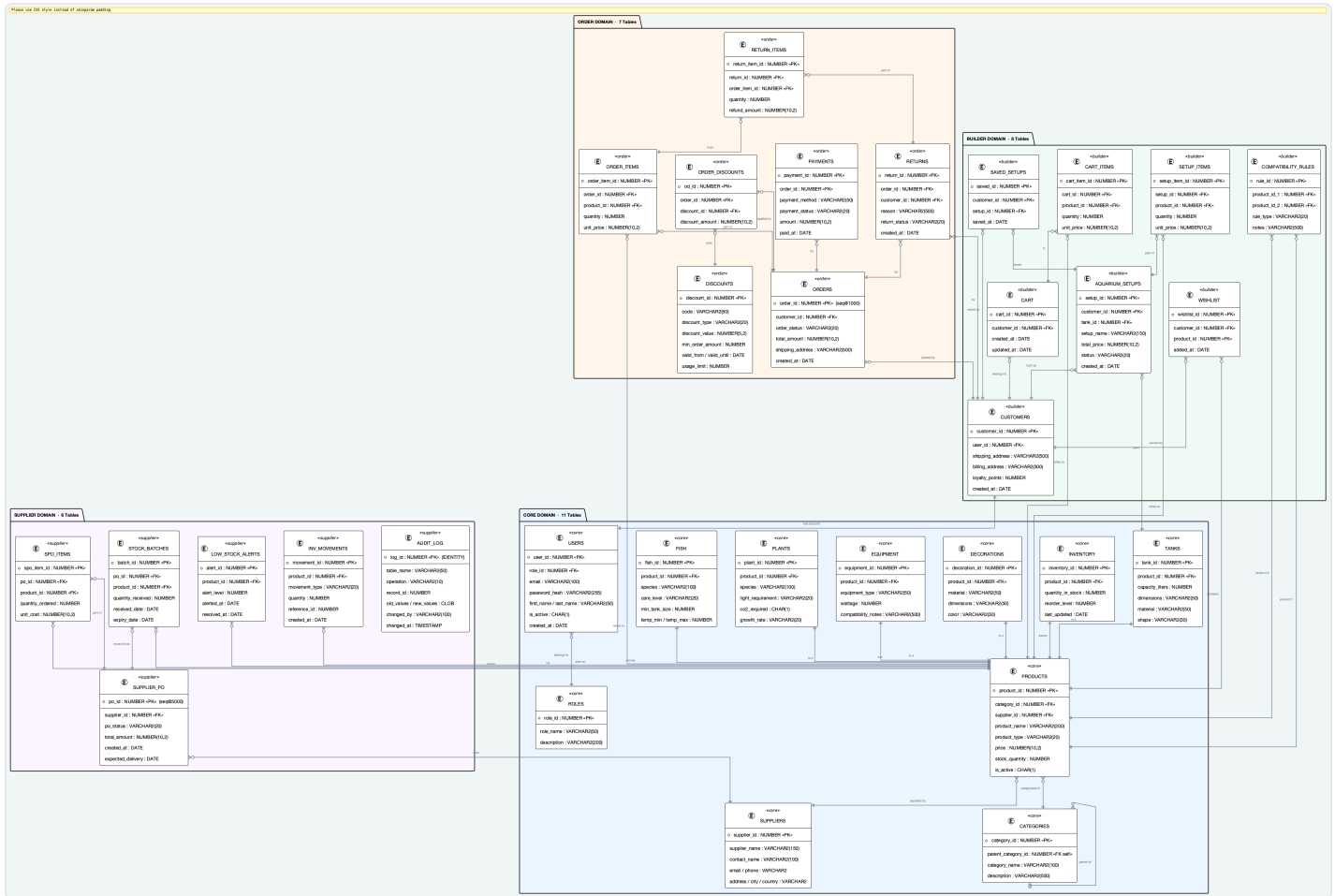
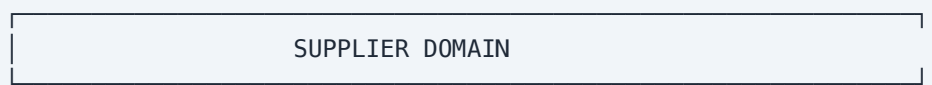
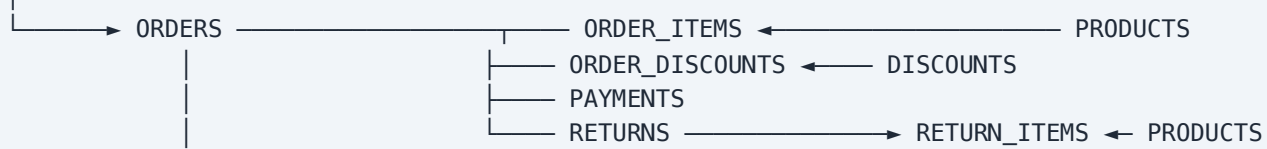
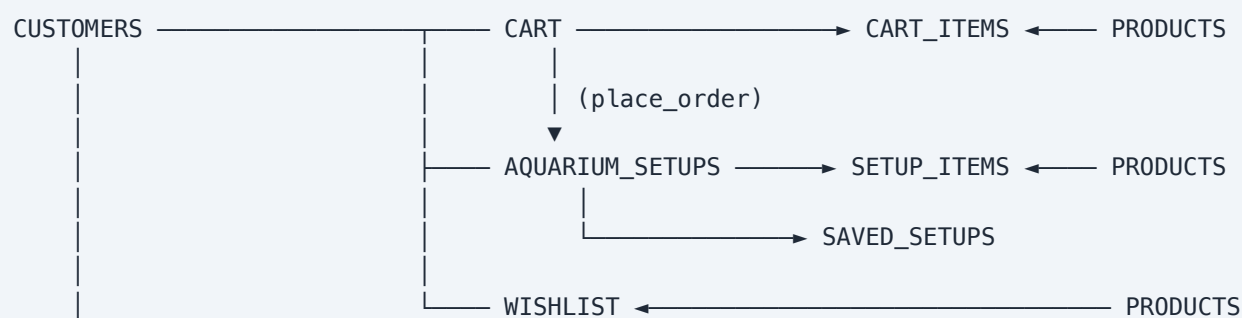
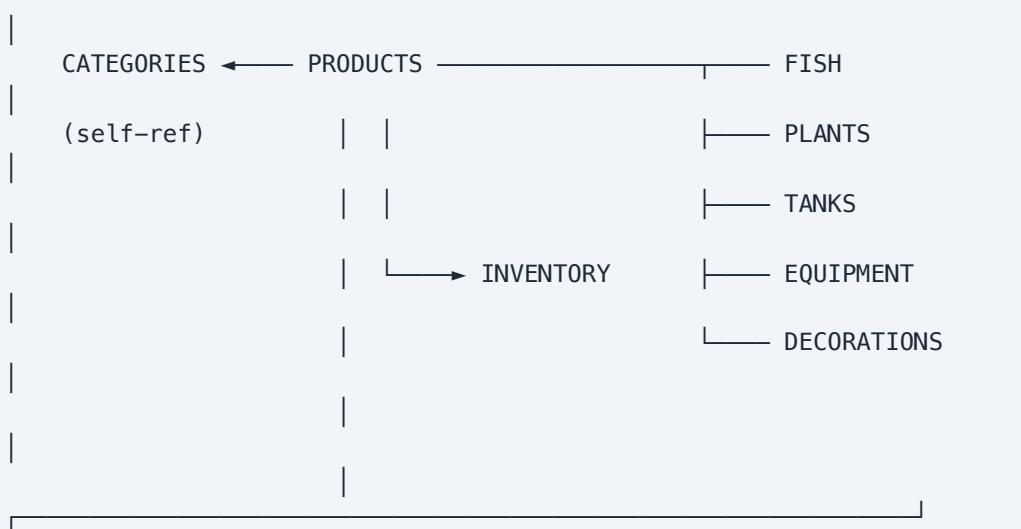
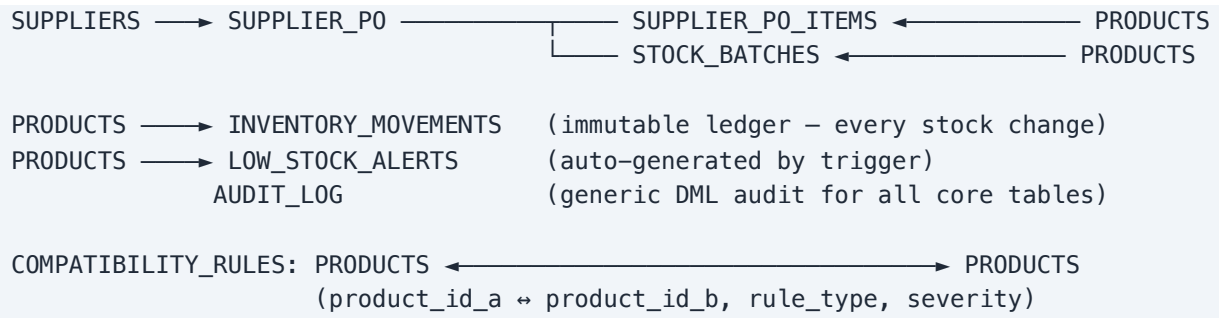


Figure 1: AquaScope ER Diagram — 32 Tables across 4 Domains (Blue: Core · Green: Builder · Orange: Order · Purple: Supplier)





7. Stored Procedures & Functions

Procedures

Procedure	Module	Description
<code>update_inventory</code>	Inventory	Central stock update — validates, adjusts <code>qty_on_hand</code> , records in movement ledger
<code>add_product_to_inventory</code>	Inventory	Creates inventory row for a new product
<code>create_aquarium_setup</code>	Builder	Creates a new DRAFT aquarium setup; returns <code>setup_id</code> OUT
<code>add_item_to_setup</code>	Builder	Adds/upserts product in setup; checks compatibility; validates <code>product_type</code> matches <code>item_type</code>
<code>save_setup</code>	Builder	Runs all 3 validations, marks setup SAVED, creates <code>saved_setups</code> record; returns <code>saved_id</code> OUT
<code>place_order</code>	Orders	Full ACID cart-to-order: stock check, discount, tax, order creation, inventory deduction, cart clear; returns <code>order_id</code> OUT
<code>cancel_order</code>	Orders	Validates cancellable status, restores inventory, writes audit
<code>update_order_status</code>	Orders	Enforces state machine transitions, writes audit
<code>create_supplier_po</code>	Supplier	Creates DRAFT PO; returns <code>po_id</code> OUT
<code>add_po_item</code>	Supplier	Adds line item to DRAFT PO; updates total
<code>submit_supplier_po</code>	Supplier	DRAFT → SUBMITTED
<code>approve_supplier_po</code>	Supplier	SUBMITTED → APPROVED
<code>receive_supplier_po</code>	Supplier	Calls <code>update_inventory</code> per item, creates batches, transitions → RECEIVED

Procedure	Module	Description
<code>process_return</code>	Orders	Approve/reject return; restocks GOOD items; issues refund payment record

Functions

Function	Returns	Description
<code>validate_tank_capacity(setup_id)</code>	0 or 1	Total fish stocking density $\leq$ tank volume
<code>validate_water_type(setup_id)</code>	0 or 1	All fish/plants match setup water_type
<code>validate_temperature(setup_id)</code>	0 or 1	All fish temp ranges overlap
<code>check_compatibility(setup_id, product_id)</code>	0 or 1	No ERROR-severity incompatible rules with existing items
<code>get_setup_total_price(setup_id)</code>	NUMBER	Sum of (quantity $\times$ unit_price) for all setup items
<code>get_customer_lifetime_value(customer_id)</code>	NUMBER	Sum of DELIVERED + SHIPPED order totals
<code>get_product_profit_margin(product_id)</code>	NUMBER	$\frac{(\text{unit_price} - \text{cost_price})}{\text{unit_price}} \times 100$
<code>is_discount_valid(code, order_total)</code>	0 or 1	Active, in-date, usage limit not exceeded, minimum order met
<code>calculate_discount_amount(code, subtotal)</code>	NUMBER	Applies PERCENT or FIXED discount

8. Triggers

Trigger	On Table	Event	Action
<code>trg_deduct_stock</code>	ORDER_ITEMS	AFTER INSERT	Calls <code>update_inventory</code> (SALE, ref_type=ORDER)
<code>trg_check_low_stock</code>	INVENTORY	AFTER UPDATE	If <code>qty_on_hand ≤ reorder_level</code> , inserts LOW_STOCK_ALERTS
<code>trg_resolve_alert</code>	INVENTORY	AFTER UPDATE	Marks matching open alerts as resolved when stock replenished
<code>trg_update_cart_ts</code>	CART_ITEMS	AFTER INSERT/UPDATE/DELETE	Updates <code>cart.updated_at</code>
<code>trg_setup_updated</code>	SETUP_ITEMS	AFTER INSERT/UPDATE/DELETE	Updates <code>aquarium_setups.updated_at</code>
<code>trg_audit_products</code>	PRODUCTS	AFTER INSERT/UPDATE/DELETE	Writes JSON old/new values to AUDIT_LOG
<code>trg_audit_orders</code>	ORDERS	AFTER INSERT/UPDATE/DELETE	Writes to AUDIT_LOG
<code>trg_audit_inventory</code>	INVENTORY	AFTER UPDATE	Writes to AUDIT_LOG
<code>trg_audit_customers</code>	CUSTOMERS	AFTER INSERT/UPDATE/DELETE	Writes to AUDIT_LOG
<code>trg_audit_users</code>	USERS	AFTER INSERT/UPDATE/DELETE	Writes to AUDIT_LOG
<code>trg_calc_order_item_total</code>	ORDER_ITEMS	BEFORE INSERT	Sets <code>line_total = quantity × unit_price</code>
<code>trg_calc_po_item_total</code>	SUPPLIER_PO_ITEMS	BEFORE INSERT/UPDATE	Sets <code>line_total = quantity_ord × unit_cost</code>
<code>trg_update_po_total</code>	SUPPLIER_PO_ITEMS	AFTER INSERT/UPDATE/DELETE	Recalculates <code>supplier_po.total_amount</code>

Trigger	On Table	Event	Action
trg_reserve_stock	CART_ITEMS	AFTER INSERT	Reserves stock (qty_reserved += quantity)
trg_unreserve_stock	CART_ITEMS	AFTER DELETE	Releases reservation
trg_single_cart	CART	BEFORE INSERT	Enforces one cart per customer

Summary Statistics

Metric	Count
Database Tables	32
Sequences	30
Stored Procedures	14
PL/SQL Functions	9
Database Triggers	16
Database Views	8
Docker Services	3 (oracle, backend, frontend)